Physics-guided local 9B model for day-ahead PV energy

INFERENCE · AVAILABLE BEFORE TARGET DAY d

No target-day measurements in the prompt

01 Inputs

Target-day NWP: irradiance, temperature, wind, humidity
History: d−9 to d−3

02 Physics anchor

Ross cell temperature
PVWatts DC approximation
Train-fitted site / GHI bias
Output: H^{cond} (h)

03 Local 9B model

Qwen3.5-9B, 4-bit
LoRA; base as control
Prompt: site, NWP, history
Physics anchor H^{cond}

04 Decode

A / B / C token logits
p_k: restricted softmax
m_k: train-class medians

$$\hat{r} = \sum_k p_k m_k$$

05 Daily forecast

$\hat{H} = \text{clip}(H^{\text{cond}} + \hat{r}, 0, 24)$
$\hat{E} = C\hat{H}$ (MWh)
Next-day site energy

TRAINING & EVALUATION

Observed AC power

Training-period actuals
$H^{\text{actual}} = E^{\text{actual}}/C$
Never in inference prompt

Train-only residual labels

$r = H^{\text{actual}} - H^{\text{cond}}$
Train terciles → A / B / C
Train class medians → m_k

Supervised adaptation

890 training station-days
Rank 8; final 2 layers
890 updates; batch size 1

Held-out scoring

Actual energy used after prediction
MAE / RMSE; date-block intervals
Base model uses no LoRA adapter

All site calibration, GHI-band offsets, class thresholds and class medians are fitted on the training split only.

Solid: inference Dashed: training / scoring